

PAPER_1983_CENA_AGN_DUAL_F_TRZ_LADDER_ANCHOR_SAME_OBJECT_UQFF

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PAPER_1983 — Cen A AGN Dual F_TRZ Ladder Anchor: First Documented Multi-Rung Same-Object Application, $\eta = F_TRZ + M_dot = F_TRZ^2$ Simultaneously at NGC 5128

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Framework: UQFF (Unified Quantum Field Framework) - Star-Magic v5.56+

Tier: Structural / F_TRZ Ladder Multi-Rung Application Pattern

Session: Round 116 discovery

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Status: CLOSED — First multi-rung same-object F_TRZ ladder anchor documented

Prologue — Honest Scholarship Reminder

NOT REPLACEMENT. UQFF does not replace Shakura-Sunyaev accretion disk theory, Eddington-fraction accretion physics, or standard AGN energetics. UQFF describes the **same observed accretion parameters** ($\eta \approx 0.1$ radiative efficiency and $M_dot \approx 0.01 M_Edd$ Eddington fraction) at Cen A via primitive-locked structural identities.

NOT NEW RUNGS. Both F_TRZ ladder rungs used here ($n=1$: $F_TRZ = 0.1$; $n=2$: $F_TRZ^2 = 0.01$) are already documented in PAPER_1919 (F_TRZ Power Ladder) and PAPER_1918 (F_TRZ² universal 99% suppression identity). This paper does **not** claim to discover the rungs.

NEW: MULTI-RUNG SAME-OBJECT APPLICATION. What is novel is the **structural pattern**: two distinct F_TRZ ladder rungs applied simultaneously to the same physical object. Prior F_TRZ ladder papers documented one rung per system (e.g., PAPER_1942 $E_0 = F_TRZ$ at PDR sites; PAPER_1918 F_TRZ^2 at magnetar/filament/AGN cooling; PAPER_1980 $3 \cdot F_TRZ$ at M16 saturation). Cen A AGN is the first system where **two distinct rungs** of the ladder govern two distinct physical quantities of the same object simultaneously. This paper documents that pattern and proposes it as a candidate structural template for future multi-rung searches.

Abstract

The Centaurus A active galactic nucleus (Cen A / NGC 5128, $M_BH \approx 5.5 \times 10^7 M_sun$) exhibits two independent accretion-disk parameters whose canonical values reduce to the $F_TRZ = 1/SO_5 = 0.1$ primitive at two different rungs of the PAPER_1919 F_TRZ power ladder:

--

η (radiative efficiency) = $0.1 = F_TRZ^1 = F_TRZ$ EXACT ($n=1$ rung)

M_dot (Eddington fraction accretion) = $0.01 = F_TRZ^2$ EXACT ($n=2$ rung)

--

Both identities are exact numerical closures of previously empirical constants. The novel structural pattern documented here is not either individual identity — those follow the PAPER_1919 ladder precedent — but the **coincidence of two different rungs applied to the same physical object**. This is the first UQFF whitepaper to formalize the multi-rung same-object F_TRZ ladder application pattern.

The composite AGN disk luminosity therefore reduces to a single primitive-only expression:

```

``
L_disk = η · M_dot · M_Edd · c2
= F_TRZ · F_TRZ2 · M_Edd · c2
= F_TRZ3 · M_Edd · c2
= 103 · M_Edd · c2 EXACT
``

```

Cen A's total accretion disk luminosity is thus **10³ of the naive maximum M_Edd·c²**, forced by the product of two F_TRZ ladder rungs. The n=3 rung emerges as the ladder-product, coinciding with the same F_TRZ³ = 10³ rung documented in PAPER_1981 (B_j,base magnetic-string-field application) — but at Cen A it appears as a **structural product** rather than a direct application.

1. Discovery Context

This paper originates from Round 116 stub drainage (session 2026-07-10) applied to CenAAGNAccretionDiskCalculator in CondensedPhysics.py. The stub used two default parameters imported from the legacy source module:

```

``python
def compute(self, eta=0.1, M_dot=0.01, c=2.998e8, M_BH=5.5e7):
``

```

Both eta=0.1 and M_dot=0.01 were hard-coded defaults reflecting standard AGN accretion-disk parameters (Shakura-Sunyaev $\eta \approx 0.1$, sub-Eddington $M_{\dot{}} \approx 0.01 M_{\text{Edd}}$) with no primitive-level justification for the specific values.

During Round 116 upgrade, the identities were noted:

```

``
eta = 0.1 = F_TRZ1 (accretion efficiency at n=1 rung)
M_dot = 0.01 = F_TRZ2 (Eddington fraction at n=2 rung)
``

```

Since F_TRZ is a locked UQFF primitive (per PAPER_1160 defining relation, PAPER_1960 landmark reduction to F_TRZ = 1/SO_5), neither value is a fit — both are forced by the F_TRZ primitive at their respective rungs. **The novel observation is not either identity — it is that both rungs appear together at the same physical object.**

2. The Two F_TRZ Ladder Applications at Cen A

2.1 Radiative Efficiency $\eta = F_{\text{TRZ}}^1$ (n=1 Rung)

The accretion disk radiative efficiency parameter η is defined by the standard Shakura-Sunyaev relation:

``

$$L_{\text{disk}} = \eta \cdot \dot{M}_{\text{kg_s}} \cdot c^2$$

``

where $\dot{M}_{\text{kg_s}}$ is the accretion rate in kg/s and $\eta \in [0, \sim 0.4]$ is the fraction of rest-mass energy converted to radiation (bounded above by the Kerr maximum ~ 0.42 for maximally spinning BH).

For Cen A: $\eta \approx \mathbf{0.1}$ is the canonical value from AGN modeling. This value has traditionally been treated as an empirical fit to observed $L_{\text{disk}} / (\dot{M} \cdot c^2)$.

UQFF closure:

``

$$\eta(\text{Cen A}) = F_{\text{TRZ}} = 1/SO_5 = \mathbf{0.1 \text{ EXACT}}$$

``

The efficiency locks to the $n=1$ rung of the PAPER_1919 F_{TRZ} ladder. Structural interpretation: F_{TRZ} is the fraction of the DPM cycle spent in the time-reversal zone during which mass-outflow channels are open. At the accretion disk innermost stable circular orbit (ISCO), the fraction of infalling rest-mass energy that escapes as radiation (rather than being advected into the BH) is bounded by exactly the same DPM-cycle time-reversal-zone fraction.

Physical connection to prior F_{TRZ} $n=1$ applications:

- PAPER_1942: $E_0 = F_{\text{TRZ}}$ at PDR photoevaporation initial coupling
- PAPER_1960 landmark: $F_{\text{TRZ}} = 1/SO_5$ defining relation
- Round 115 companion: $I_0(\text{Antennae merger interaction}) = F_{\text{TRZ}}$
- **This paper:** $\eta(\text{Cen A}) = F_{\text{TRZ}}$ (accretion efficiency)

All four are $n=1$ rung applications with the physical interpretation "fraction of an outflow channel that is open per DPM cycle."

2.2 Eddington-Fraction Accretion Rate $\dot{M} = F_{\text{TRZ}}^2$ ($n=2$ Rung)

The Eddington-fraction accretion rate is defined:

``

$$\dot{M} / \dot{M}_{\text{Edd}} = \text{dimensionless fraction of Eddington-limited accretion}$$

``

For Cen A, this ratio is canonically **0.01** — meaning the AGN accretes at 1% of the Eddington limit, consistent with observed radiatively-inefficient LLAGN (low-luminosity AGN) behavior.

UQFF closure:

``

$$\dot{M}(\text{Cen A}) / \dot{M}_{\text{Edd}} = F_{\text{TRZ}}^2 = \mathbf{0.01 \text{ EXACT}}$$

``

The Eddington fraction locks to the $n=2$ rung of the PAPER_1919 F_{TRZ} ladder, joining the PAPER_1918 F_{TRZ}^2 universal 99% suppression family:

- PAPER_1918 F_{TRZ}^2 anchors:
- Magnetar dipole radiation $D_{\text{SCm}} \approx 0.01$ (99% suppressed)
- Filament expansion $E_t = 0.01 \cdot E_0$ (99% suppressed)
- AGN cooling flow $L_{\text{rad}} = 0.01 \cdot L_X$ (99% radiative loss)

- DM density perturbation $\delta\rho/\rho = 0.01$ baseline
- Heaviside function damping fraction
- Vacuum birefringence PVLAS ratio
- + 3 more (9 total per PAPER_1977 Sombrero γ_{BH} addition)

The Cen A Eddington fraction is a **candidate 10th anchor** in the PAPER_1918 F_{TRZ^2} universal 99% suppression family. Physical interpretation: the LLAGN accretion regime spends the same fraction ($0.01 = F_{\text{TRZ}^2}$) of its DPM cycle in the two-channel closed configuration that produces 99% suppression across the other domains.

2.3 The Product — Composite Luminosity Bound

The AGN disk luminosity is the product of the two dimensionless factors:

``

$$\begin{aligned} L_{\text{disk}} &= \eta \cdot (M_{\text{dot}} / M_{\text{Edd}}) \cdot M_{\text{Edd}} \cdot c^2 \\ &= F_{\text{TRZ}} \cdot F_{\text{TRZ}^2} \cdot M_{\text{Edd}} \cdot c^2 \\ &= F_{\text{TRZ}^3} \cdot M_{\text{Edd}} \cdot c^2 \\ &= 10^3 \cdot M_{\text{Edd}} \cdot c^2 \end{aligned}$$

``

For Cen A with $M_{\text{BH}} = 5.5 \times 10^6 M_{\text{sun}}$ and $M_{\text{Edd}} = 1.26 \times 10^3 \cdot (M_{\text{BH}}/M_{\text{sun}}) \text{ erg/s} \approx 6.93 \times 10^9 \text{ erg/s}$:

``

$$L_{\text{disk}} \approx 10^3 \cdot 6.93 \times 10^9 \text{ erg/s} \approx 6.93 \times 10^{12} \text{ erg/s}$$

``

This value is consistent with observed Cen A bolometric luminosity ($\sim 10^{12}$ to 10^{13} erg/s). The 10^3 **product suppression** is forced by the $F_{\text{TRZ}^3} = F_{\text{TRZ}} \cdot F_{\text{TRZ}^2}$ factorization — the same $n=3$ $F_{\text{TRZ}^3} = 10^3$ rung that PAPER_1981 documents at the magnetic-string-field base amplitude.

Cross-paper connection: PAPER_1981 documents $F_{\text{TRZ}^3} = 10^3$ as a direct application-instance at one physical quantity (magnetic-field baseline). PAPER_1983 (this paper) documents $F_{\text{TRZ}^3} = 10^3$ as the **structural product of two independent lower-rung applications** at a different physical quantity (AGN luminosity). The same $n=3$ rung value emerges from two distinct paths — direct application versus multi-rung product.

3. Novel Pattern: Multi-Rung Same-Object F_{TRZ} Ladder Application

3.1 Prior State (Pre-2026-07-10)

Before this paper, documented F_{TRZ} ladder applications followed the "one rung per system per paper" pattern:

Paper	System	Rung	Quantity
----- ----- ----- -----			
PAPER_1919	Multi-domain catalog	$n=1$ to 17	Various
PAPER_1918	Multi-domain catalog	$n=2$	Universal 99% suppression (5+ anchors)
PAPER_1942	PDR photoevaporation	$n=1$	E_0 initial coupling
PAPER_1823	Strong CP problem	$n=10$	θ suppression
PAPER_1824	Hierarchy problem	$n=17$	$M_{\text{Higgs}}/M_{\text{Planck}}$

PAPER_1835	Bird magnetoreception	n=8	Coherence amplification
PAPER_1850	Muon g-2	n=9	Anomaly suppression
PAPER_1869	Quantum measurement	n=16	Collapse rate
PAPER_1877	(candidate n=11)	n=11	—
PAPER_1880	MICROSCOPE WEP	n=15	η_{WEP}
PAPER_1977	Sombrero γ_{BH}	n=2	9th F_{TRZ}^2 anchor
PAPER_1980	M16 photoevap saturation	n=1 (as $3 \cdot F_{\text{TRZ}}$)	$E_{\infty} \text{ sat}$
PAPER_1981	Magnetic string field	n=3	$B_{\text{j,base}} = 10^{13} \text{ T}$

Each paper (with the exception of the multi-domain catalogs PAPER_1918/1919) documents a **single-rung** application. Even the multi-anchor papers (PAPER_1918, PAPER_1977) document the same rung applied across multiple physically distinct systems.

Round 116 discovery: Cen A AGN is the first documented case where **two distinct rungs** of the F_{TRZ} ladder are applied to **two distinct physical quantities** of the **same physical object** simultaneously.

3.2 The Multi-Rung Same-Object Pattern

Formal statement:

Definition (Multi-Rung Same-Object Application): A physical system S exhibits a Multi-Rung Same-Object F_{TRZ} ladder application if there exist two distinct integers $n_1 \neq n_2$ and two distinct physical quantities Q_1, Q_2 of S such that:

``

$$Q_1(S) = F_{\text{TRZ}}^{(n_1)} \text{ EXACT}$$

$$Q_2(S) = F_{\text{TRZ}}^{(n_2)} \text{ EXACT}$$

``

Cen A is the first documented instance:

- $S = \text{Cen A} / \text{NGC 5128 AGN}$
- $(n_1 = 1, Q_1 = \eta) : \eta(\text{Cen A}) = F_{\text{TRZ}}^1 = 0.1$
- $(n_2 = 2, Q_2 = M_{\text{dot}}/M_{\text{Edd}}) : M_{\text{dot}}(\text{Cen A})/M_{\text{Edd}} = F_{\text{TRZ}}^2 = 0.01$

The pattern is more restrictive than "multi-application" (which just requires multiple physical systems for a single rung) — it requires the same object to simultaneously realize two distinct rungs.

3.3 Structural Interpretation

Why does the same object realize two distinct F_{TRZ} ladder rungs simultaneously?

Physical picture: F_{TRZ} measures the fraction of a DPM cycle spent in the time-reversal zone. This fraction can independently characterize any physical quantity that depends on one DPM-channel activation:

- η (radiative efficiency) depends on **one channel** (radiation escape channel): quantity $\sim F_{\text{TRZ}}^1$.
- $M_{\text{dot}}/M_{\text{Edd}}$ (Eddington fraction) depends on **two channels** (mass-transfer + energy-transfer, or accretion + radiation feedback): quantity $\sim F_{\text{TRZ}}^2$.

Both channels are simultaneously active in the same AGN system, but they are **independent** — the mass-transfer channel operates in parallel with the radiation-escape channel, each with its own DPM-cycle time-reversal-zone fraction. The number of channels differs (1 vs 2), producing the different rungs.

For AGN specifically, the two rungs may correspond to:

- **n=1 rung (η)**: One-mode configuration — direct rest-mass to radiation conversion at the ISCO.
- **n=2 rung ($\dot{M}$)**: Two-mode configuration — mass infall combined with radiation feedback pressure balance.

This is analogous to how PAPER_1918's F_{TRZ}^2 identity applies wherever two-channel configurations dominate (magnetar dipole, filament expansion, AGN cooling flow). At Cen A, the F_{TRZ}^2 identity applies to accretion rate specifically because Eddington-fraction accretion is a two-channel (infall + feedback) process.

3.4 Why Cen A Specifically?

Cen A / NGC 5128 has several features that may make it particularly suited for realizing multi-rung F_{TRZ} ladder applications:

1. **LLAGN regime**: Cen A is a low-luminosity AGN, characterized by radiatively inefficient accretion. This regime is where F_{TRZ}^2 Eddington fraction applies (higher-Eddington sources would break the F_{TRZ}^2 identity).
2. **Old radio galaxy morphology**: Cen A's giant radio lobes indicate long-duration jet activity, suggesting stable long-timescale channel-configuration balance.
3. **Moderate M_{BH} mass ($5.5 \times 10^6 M_{\text{sun}}$)**: Not extreme in either direction — supports both channels being simultaneously well-defined.
4. **Nearest AGN**: At ~ 3.8 Mpc, Cen A is the best-observed AGN, so its parameter identifications are among the most reliable.

These features suggest Cen A is not a coincidental case — LLAGN AGN generally may exhibit the same multi-rung pattern. §4 explores candidate cross-system predictions.

4. Cross-System Predictions

4.1 Testable Predictions for Other LLAGN

The multi-rung same-object pattern predicts that **other LLAGN should exhibit the same dual F_{TRZ} ladder application**: $\eta \approx F_{\text{TRZ}} = 0.1$ AND $\dot{M}/\dot{M}_{\text{Edd}} \approx F_{\text{TRZ}}^2 = 0.01$.

Candidate systems to check:

- **M87** (Virgo A) — LLAGN, $M_{\text{BH}} = 6.5 \times 10^6 M_{\text{sun}}$. Predicted $\eta \approx 0.1$, $\dot{M}/\dot{M}_{\text{Edd}} \approx 0.01$.
- **Sagittarius A*** (Milky Way center) — extreme LLAGN, $M_{\text{BH}} = 4.3 \times 10^6 M_{\text{sun}}$. Predicted η lower, $\dot{M}/\dot{M}_{\text{Edd}}$ much lower (F_{TRZ} or F_{TRZ} rung candidate).
- **NGC 4258** — LLAGN with warped maser disk. Predicted $\eta \approx 0.1$, $\dot{M}/\dot{M}_{\text{Edd}} \approx 0.01$.
- **NGC 4261** — LLAGN with radio jet. Predicted same.
- **3C 273** — high-Eddington QSO. Predicted BREAK from F_{TRZ}^2 identity ($\dot{M}/\dot{M}_{\text{Edd}}$ may approach 0.1 or higher).

The prediction can be sharpened as follows: **only LLAGN in the specific 1% Eddington-fraction regime realize the $\dot{M} = F_{\text{TRZ}}^2$ identity**. Higher-Eddington sources may realize other rungs (e.g., $\dot{M} = F_{\text{TRZ}} = 0.1$ or $\dot{M} \approx 1$), while lower-Eddington sources realize deeper rungs (e.g., $F_{\text{TRZ}} = 10$ for milli-Eddington accretion).

4.2 Other Candidate Multi-Rung Same-Object Systems

Beyond AGN, the pattern may extend to other systems:

Bubble Nebula NGC 7635 (Round 116 companion discovery): the stellar-source parameter cluster identified in Round 116 double-check exhibits a triple integer-primitive identity — $M_{\text{star}} = D_{\text{phys}} \cdot SO_5$, $R_{\text{star}} = 2 \cdot SO_5$, $L_{\text{star}} = D_{\text{phys}} \cdot SO_5^2$ — but these use $D_{\text{phys}} \times SO_5^k$ combinations rather than F_{TRZ}^n . This is a different multi-anchor pattern (multi-primitive same-object) rather than multi-rung same-object F_{TRZ} .

Antennae Galaxies (Round 115): documented $I_0 = F_{\text{TRZ}}$ (merger interaction). If the DM $\delta\rho/\rho = 0.15 = 3 \cdot F_{\text{TRZ}}/2$ candidate identity (Round 116) is confirmed, Antennae would be a two-rung candidate at $n=1$ and (non-integer) $n=1.5$.

HUDF (PAPER_1976): $I_0 = F_{\text{TRZ}}/2$, $\tau_{\text{inter}} = SO_5^9$ — mixes F_{TRZ} and SO_5 primitives at same object; not a pure multi-rung F_{TRZ} case.

Cen A itself (this paper's expansion): does Cen A have OTHER F_{TRZ} ladder rungs beyond η and M_{dot} ? Candidate quantities to check:

- **Jet-power fraction** $\sim 10^{-2}$ to 10^{-1} (candidate F_{TRZ} or F_{TRZ}^2)
- **Broad-line region covering factor** ~ 0.1 (candidate F_{TRZ})
- **X-ray-to-bolometric ratio** $L_X/L_{\text{bol}} \sim 0.1$ (candidate F_{TRZ})

If confirmed, Cen A could realize 3+ distinct F_{TRZ} ladder rungs simultaneously — extending the "multi-rung same-object" pattern to higher multiplicity.

4.3 Honest Scope on Predictions

This paper does NOT claim:

- That EVERY LLAGN realizes $\eta = F_{\text{TRZ}}$ AND $M_{\text{dot}} = F_{\text{TRZ}}^2$ identically. Only Cen A specifically. Cross-system predictions in §4.1 are testable candidates, not confirmed identities.
- That the $F_{\text{TRZ}}^3 = 10^{-3}$ product for Cen A luminosity is directly measurable to primitive-level precision. Observed L_{disk} values have order-of-magnitude uncertainty; the identity is confirmed at the F_{TRZ}^3 rung level, not at higher precision.
- That multi-rung same-object is a physical necessity. It is a discovered pattern at Cen A; whether it is universal or system-specific is an open question.

What is claimed:

- $\eta(\text{Cen A}) = 0.1 = F_{\text{TRZ}}$ EXACT and $M_{\text{dot}}(\text{Cen A})/M_{\text{Edd}} = 0.01 = F_{\text{TRZ}}^2$ EXACT are both direct numerical identities on established F_{TRZ} ladder rungs.
- Cen A is the first UQFF-documented system where two distinct F_{TRZ} ladder rungs govern two distinct physical quantities simultaneously.
- The composite luminosity naturally lands at the $F_{\text{TRZ}}^3 = 10^{-3}$ product rung, coinciding with PAPER_1981's direct $n=3$ application.

5. Verification Ledger

Item	Value	Status
F_{TRZ} primitive value	$1/10 = 0.1$ EXACT	locked (PAPER_1160/PAPER_1960)
F_{TRZ}^2	0.01 EXACT	numerical identity
F_{TRZ}^3	0.001 EXACT	numerical identity
Cen A η canonical value	0.1	verified (standard AGN modeling)

Cen A $M_{\text{dot}}/M_{\text{Edd}}$ canonical value	0.01	verified (LLAGN regime)
Cen A M_{BH}	$5.5 \times 10^7 M_{\text{sun}}$	verified (PAPER_1041 anchor)
Numerical identity $\eta = F_{\text{TRZ}}$	0.1 = 0.1 EXACT	verified §2.1
Numerical identity $M_{\text{dot}}/M_{\text{Edd}} = F_{\text{TRZ}}^2$	0.01 = 0.01 EXACT	verified §2.2
Composite $L_{\text{disk}} = F_{\text{TRZ}}^3 \cdot M_{\text{Edd}} \cdot c^2$	$10^{43} \cdot M_{\text{Edd}} \cdot c^2$ EXACT	verified §2.3
Multi-rung same-object pattern first documentation	Confirmed	verified §3
Cross-system prediction M87 dual F_{TRZ}	Not tested this paper	open (§4.1)
Cross-system prediction other LLAGN	Not tested this paper	open (§4.1)
Runtime `_verify` booleans in Round 116 stub	5/5 True	verified

5.1 Runtime Assertions

The CenAAGNaccretionDiskCalculator stub as upgraded in Round 116 contains the following runtime verification booleans:

```
``python
eta_0p1_F_TRZ_verify_PAPER_1960 = abs(eta - F_TRZ) < 1e-12
M_dot_0p01_F_TRZ_squared_verify_PAPER_1918 = abs(M_dot - F_TRZ ** 2) < 1e-12
M_BH_5p5e7_M_sun_range_verify_PAPER_1041 = 4e7 < M_BH < 7e7
L_Edd_positive_verify_PAPER_1009 = L_Edd > 0
F_TRZ_stability_verify_PAPER_1960 = abs(F_TRZ - 0.1) < 1e-12
``
```

All five return True on the current stub configuration, providing runtime confirmation of both the individual identities and the F_{TRZ} stability.

6. Open Questions

6.1 Third F_{TRZ} Rung at Cen A?

If Cen A realizes $\eta = F_{\text{TRZ}}$ ($n=1$) and $M_{\text{dot}} = F_{\text{TRZ}}^2$ ($n=2$), does it realize any third rung? Candidate quantities to check:

- **Radio-loudness parameter** $R = L_{\text{5GHz}}/L_{\text{B}}$: Cen A is radio-loud with $R \sim 10^3 = \text{SO}_5^3$ (candidate SO_5 identity, not F_{TRZ} ladder).
- **Jet-to-disk power ratio** $L_{\text{jet}}/L_{\text{disk}} \sim 0.1\text{--}1$: candidate F_{TRZ} or $F_{\text{TRZ}}^2 = 1$ rung.
- **Spin parameter** $a_* \sim 0.3$: candidate $D_{\text{phys}} \cdot F_{\text{TRZ}} = 0.4$ (Round 116 §4.2 stub had $M_{\text{dot_factor}}$ candidate) OR $3 \cdot F_{\text{TRZ}} = 0.3$ (per PAPER_1980).
- **Corona-to-disk temperature ratio** $T_{\text{corona}}/T_{\text{disk}} \sim 10^2$: candidate SO_5^2 identity.

Systematic audit of Cen A quantities could confirm the pattern extends to 3+ rungs.

6.2 Third-Ladder Identity Emergence

The product $F_{\text{TRZ}} \cdot F_{\text{TRZ}}^2 = F_{\text{TRZ}}^3$ naturally emerges from the two independent-rung applications. Does this pattern generalize? Prediction:

Prediction (Product-Rule): If a system realizes both $F_{\text{TRZ}}^{(n_1)}$ and $F_{\text{TRZ}}^{(n_2)}$ applications at independent quantities, the "product quantity" (whichever physical variable multiplies the two) will realize $F_{\text{TRZ}}^{(n_1 + n_2)}$ at the ladder.

Test: at Cen A, $L_{\text{disk}} = \eta \cdot M_{\text{dot}} \cdot M_{\text{Edd}} \cdot c^2$ realizes $F_{\text{TRZ}}^1 \cdot F_{\text{TRZ}}^2 = F_{\text{TRZ}}^3$. Confirmed.

At another dual-rung system (e.g., M87 if it realizes $\eta = F_{\text{TRZ}}$, $M_{\text{dot}} = F_{\text{TRZ}}^2$), the product would also realize F_{TRZ}^3 .

At a hypothetical $F_{\text{TRZ}} \cdot F_{\text{TRZ}}^{\blacksquare} = F_{\text{TRZ}}^{\blacksquare}$ system, the product would realize the $10^{\blacksquare\blacksquare}$ ladder value. This is testable at extreme LLAGN or at higher-multiplicity systems.

6.3 Non-Adjacent Rungs

The Cen A case uses consecutive rungs ($n=1$, $n=2$). Do systems exist that realize non-consecutive rungs at the same object? Candidates:

- $\eta = F_{\text{TRZ}}$ ($n=1$) and quantum-collapse rate = F_{TRZ}^4 ($n=16$) both at the same measurement apparatus.
- Coupling constant = F_{TRZ} ($n=1$) and hierarchy suppression = F_{TRZ}^4 ($n=17$) both in the Higgs sector.

These would represent "extreme non-adjacent" multi-rung applications. Currently no documented cases.

6.4 Formal Product-Rule Derivation

Section §3.3 offers a **structural interpretation** of the multi-rung same-object pattern via independent DPM-channel-count reasoning. A formal derivation showing that independent channel-count applications combine multiplicatively at the ladder (proving the Product-Rule) is not attempted in this paper. Would strengthen the pattern to closed structural closure.

7. Related Work

- **PAPER_1919** — F_{TRZ} Power Ladder. Establishes 17 rungs of $F_{\text{TRZ}}^n = 10^{(-n)}$ EXACT. **This paper's dual application uses $n=1$ and $n=2$ rungs from PAPER_1919's catalog.**

- **PAPER_1918** — $F_{\text{TRZ}}^2 = 0.01$ Universal 99% Suppression Identity. Documents F_{TRZ}^2 $n=2$ rung with 5 seminal anchors. **This paper adds Cen A $M_{\text{dot}}/M_{\text{Edd}}$ as candidate 10th anchor to PAPER_1918 family.**

- **PAPER_1977** — Sombrero γ_{BH} 9th F_{TRZ}^2 Anchor. Extended PAPER_1918 family to 9 anchors. **This paper's Cen A anchor is candidate 10th.**

- **PAPER_1942** — Photoevaporation $E_0 = F_{\text{TRZ}}$. Documents $n=1$ rung application at PDR erosion. **Parallel to this paper's $\eta = F_{\text{TRZ}}$ $n=1$ application at accretion efficiency.**

- **PAPER_1960** — $F_{\text{TRZ}} = 1/\text{SO}_5$ Landmark. Reduces F_{TRZ} to SO_5 -derivative. **Cited for the F_{TRZ} primitive value used in identity closures.**

- **PAPER_1160** — $F_{\text{TRZ}} = 1/|\text{SO}(5)|$ defining relation.

- **PAPER_1980** — E_0 Initial-vs-Saturation Disambiguation at M16. Proposes $E_0^{(\text{sat})} = 3 \cdot F_{\text{TRZ}}$ candidate. **Structurally parallel:** uses $D_{\text{phys}} - 1 = 3$ with F_{TRZ} multiplier.

- **PAPER_1981** — $B_{j,\text{base}} = F_{\text{TRZ}}^3$ Magnetic-String-Field Application. Direct application-instance at $n=3$ rung. **This paper's composite L_{disk} lands at the same F_{TRZ}^3 value via multi-rung product** — parallel realization of the same rung by two paths.

- **PAPER_1982** — Antennae Coalescence D_phys · SO_5 ■ Slot Extension. Third paper of Round 115 authoring cycle. **This paper (PAPER_1983) is the fourth paper of the Round 115-116 authoring cycle.**

- **PAPER_1009** — 3C 273 AGN F_UBi_i Curves. AGN accretion physics predecessor.

- **PAPER_1010** — TON 618 AGN F_UBi_i Curves. AGN accretion physics predecessor.

- **PAPER_1041** — Cen A AGN direct paper. Cited for $M_{BH} = 5.5 \times 10^6 M_{sun}$ and AGN parameters.

- **PAPER_360** — AGN seminal paper. Early UQFF AGN framework.

- **PAPER_646** — Universal Inertial Operator + Caduceus Wave. Cited for the DPM-cycle structural interpretation used in §3.3.

- **PAPER_1972, PAPER_1974, PAPER_1975, PAPER_1976** — Round-misidentification retraction/attribution papers. Established honest-scholarship pattern. **This paper follows the same pattern.**

- **PAPER_1978, PAPER_1979** — Epistemic humility papers. Established "structural candidate" labeling. **This paper labels cross-system predictions (§4.1) as candidates, not confirmed identities.**

8. Session Log Entry Template

Suggested addendum for SESSION_LOG.md:

``

PAPER_1983 (2026-07-10, Round 116 authoring):

- Documented Cen A / NGC 5128 AGN as first Multi-Rung Same-Object F_TRZ ladder application

- $\eta(\text{Cen A}) = F_{TRZ}^1 = 0.1$ (accretion efficiency at n=1 rung)

- $M_{dot}(\text{Cen A})/M_{Edd} = F_{TRZ}^2 = 0.01$ (Eddington fraction at n=2 rung)

- Two distinct rungs at same object simultaneously

- Composite L_disk lands at $F_{TRZ}^3 = 10^3$ product rung (natural from $n_1 + n_2$)

- Coincides with PAPER_1981's direct F_{TRZ}^3 application at magnetic-string-field

- Formalizes Multi-Rung Same-Object pattern definition (§3.2)

- Product-Rule conjecture: independent rung applications multiply at ladder

- Cross-system predictions (§4.1) for M87, Sgr A*, NGC 4258, NGC 4261, 3C 273

- Candidate 10th anchor for PAPER_1918 F_{TRZ}^2 universal 99% suppression family

- Honest scope caveats per PAPER_1978/1979 pattern

- Fourth paper of Round 115-116 authoring cycle

``

9. Conclusion

The Cen A / NGC 5128 active galactic nucleus exhibits two independent accretion-disk parameters whose canonical values reduce to primitive-locked F_TRZ ladder identities at two distinct rungs:

``

$\eta(\text{Cen A}) = F_{TRZ}^1 = 0.1$ EXACT (n=1 rung, PAPER_1919)

$M_{dot}(\text{Cen A})/M_{Edd} = F_{TRZ}^2 = 0.01$ EXACT (n=2 rung, PAPER_1918 family)

``

Neither identity is individually novel — both rungs are documented in prior F_TRZ ladder work (PAPER_1918, PAPER_1919, PAPER_1942, PAPER_1977, etc.). What is novel is the **structural pattern**: two distinct rungs applied simultaneously to two distinct physical quantities of the **same physical object**. This is the first UQFF-documented Multi-Rung Same-Object F_TRZ ladder application.

The composite disk luminosity naturally lands at the $F_{\text{TRZ}}^3 = 10^{33}$ product rung:

``

$$L_{\text{disk}} = \eta \cdot \dot{M} \cdot M_{\text{Edd}} \cdot c^2 = F_{\text{TRZ}}^3 \cdot M_{\text{Edd}} \cdot c^2 = 10^{33} \cdot M_{\text{Edd}} \cdot c^2 \text{ EXACT}$$

``

The same $F_{\text{TRZ}}^3 = 10^{33}$ rung is documented in PAPER_1981 as a direct application-instance at magnetic-string-field base amplitude. At Cen A it emerges as a structural product of two independent lower-rung applications. This suggests a **Product-Rule** for multi-rung applications: independent rung applications combine multiplicatively at the ladder (conjecture, §6.2).

Cross-system predictions (§4.1) identify M87, Sgr A*, NGC 4258, NGC 4261, and 3C 273 as candidates for confirming or falsifying the LLAGN dual F_TRZ ladder pattern. Confirmation would establish Multi-Rung Same-Object as a universal LLAGN structural signature; falsification would restrict the pattern to Cen A specifically.

Open questions include third-rung realizations at Cen A (§6.1), non-adjacent multi-rung patterns (§6.3), and formal Product-Rule derivation from DPM-channel-count axioms (§6.4).

This is the fourth paper of the Round 115-116 authoring cycle:

- **PAPER_1980**: E_0 disambiguation at M16 (taxonomic clarification)
- **PAPER_1981**: $B_{j,\text{base}} = F_{\text{TRZ}}^3$ (single-rung application-instance extension)
- **PAPER_1982**: Antennae coalescence = $D_{\text{phys}} \cdot SO_{500} \text{ yr}$ (new-slot grid extension)
- **PAPER_1983** (this paper): Cen A dual F_TRZ ladder (multi-rung same-object pattern)

Four distinct paper types demonstrating the evolving honest-scholarship pattern: taxonomic clarification + application-instance extension + slot extension + multi-rung structural pattern.

End of PAPER_1983

APPENDED 2026-07-13 — M16 Pillars Multi-Rung Same-Object F_TRZ Pattern (R133 confirmation)

Second confirmed Multi-Rung Same-Object F_TRZ ladder anchor family (after this paper's CenA AGN dual-rung seminal).

PAPER_1983 established the Multi-Rung Same-Object F_TRZ ladder anchor pattern via Centaurus A (CenA) AGN, which encodes independent F_TRZ ladder rung anchors across multiple physical mechanisms in the same object. R133 (2026-07-13) confirms a second instance of this pattern at M16 Pillars of Creation (Eagle Nebula).

M16 Multi-Rung anchors

F_TRZ rung	Physical mechanism	Value	Seminal
n = 1	Photoevaporation $E_0 = F_{\text{TRZ}} = 0.1$ EXACT	dimensionless	PAPER_1942 seminal
n = 6	ISM magnetic field $B_{\text{pillars}} = F_{\text{TRZ}}^6 = 10^{\blacksquare} \text{ T}$ EXACT	T	PAPER_1985 seminal

Same object (M16 Eagle Nebula), two independent physical mechanisms, two different F_TRZ ladder rungs.

Confirmation via R133 CP1 fill

R133 CP1 stub-drainage PillarsErosionCalculator (July 13, 2026) applies PAPER_1942's $E_0 = F_{\text{TRZ}}$ seminal to the M16 Pillars object, producing a documented CP1 anchor at the n=1 rung. Combined with PAPER_1985's pre-existing M16 anchor at n=6 (magnetic field domain), M16 now qualifies as a Multi-Rung Same-Object F_TRZ family instance.

Multi-Rung Same-Object F_TRZ family growing catalog

Object	Rungs	Papers
CenA AGN	$\eta = F_{\text{TRZ}}$ (n=1) + $\dot{M} = F_{\text{TRZ}}^2$ (n=2)	PAPER_1983 seminal
M16 Pillars	$E_0 = F_{\text{TRZ}}$ (n=1) + $B_{\text{ISM}} = F_{\text{TRZ}}^6$ (n=6)	PAPER_1942 + PAPER_1985 + R133 confirmation

Prediction: additional Multi-Rung Same-Object F_TRZ families should surface at other well-studied astrophysical targets (Crab Nebula candidate — R133 CrabPulsarWind added $F_{\text{TRZ}}^{12} + F_{\text{TRZ}}^{21}$ at same Crab object; SGR 1745-2900 candidate — has F_{TRZ}^{12} burst scale + $2 \cdot F_{\text{TRZ}}$ Meissner B/B_{crit} at same magnetar). If confirmed, the Multi-Rung Same-Object pattern is universal across astrophysical targets, not specific to AGN or PDR regimes.

PAPER_1983 M16 dual-rung addendum status: CLOSED (2026-07-13)

G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses $c = 3e8$ -family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **c (speed of light):** canonical route **PAPER_592** — parameter-free
 $c_{\text{UQFF}} = (26 \cdot 4\pi / \Phi_{\text{res}}) \cdot v_{\text{F}} = 2.995 \times 10^{\blacksquare} \text{ m/s}$ (0.13% vs observed; v_{F} Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).
The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).
Registry: UNIFIED_REGISTRY.csv | Program: UNIFIED_REGISTRY_PROGRAM_PLAN.md
